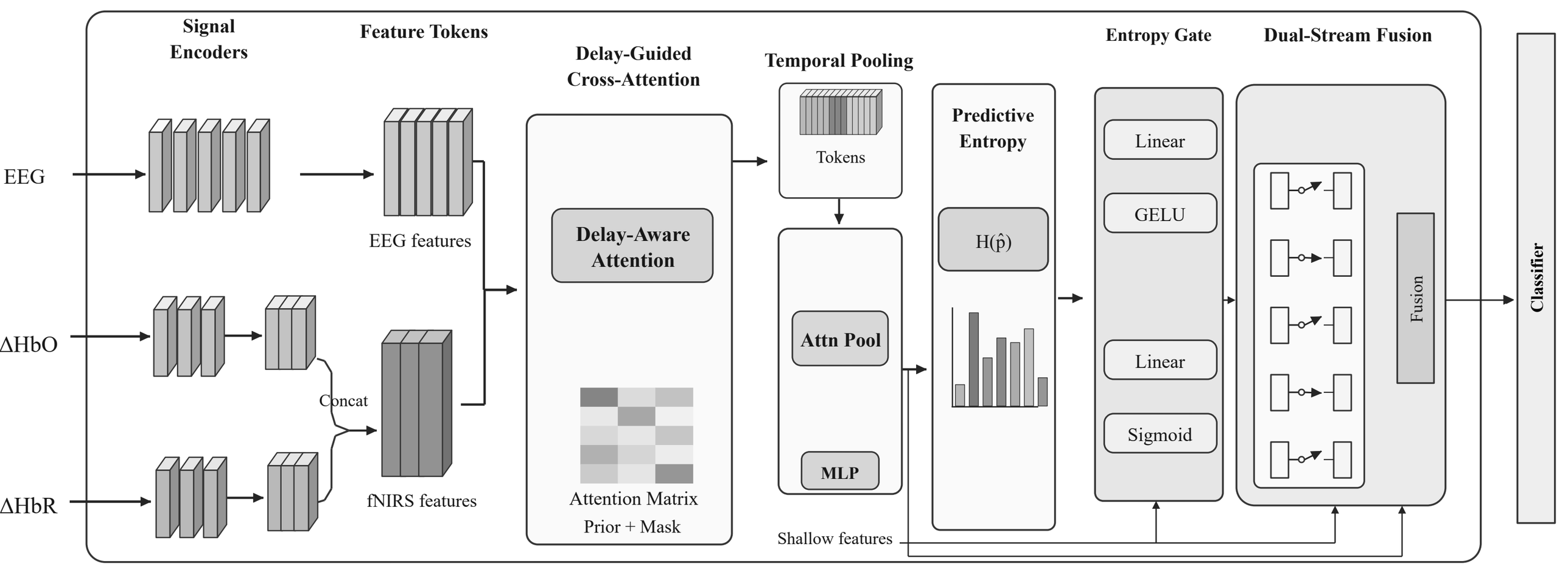

Signal Encoders
Feature Tokens
Delay-Guided Cross-Attention
Temporal Pooling
Entropy Gate
Dual-Stream Fusion
EEG
ΔHbO
ΔHbR
EEG features
Concat
fNIRS features
Delay-Aware Attention
Attention Matrix
Prior + Mask
Tokens
Attn Pool
MLP
Predictive Entropy
H(p̂)
Linear
GELU
Linear
Sigmoid
Fusion
Shallow features
Classifier